Andrew Y. K. Foong

PERSONAL WEBSITE: https://andrewfoongyk.github.io/
GOOGLE SCHOLAR: https://scholar.google.com/citations?user=2UOjgIUAAAAJ&hl=en

EXPERIENCE

MAYO CLINIC — January 2025 – Present
AI Scientist, Senior Associate Consultant & Assistant Professor of Biomedical Informatics

- Co-director of the AI & Data Analytics group in the Department of Radiation Oncology (Rochester, Minnesota). Lead research and training efforts, and translate models into clinical workflows in collaboration with physicians and operations teams.

MICROSOFT RESEARCH CAMBRIDGE — November 2022 – November 2024
Senior Researcher

- Generative modelling for structural biology in the AI4Science team. Built probabilistic generative models for protein dynamics, from formulation through implementation and large-scale evaluation on Azure. Contributed to BioEmu, culminating in a co-first-author paper in *Science*.

GOOGLE DEEPMIND — February 2022 – May 2022
Research Scientist Intern

- Data-efficient and Bayesian learning under Dr. Michalis Titsias. Developed deep generative models that combine variational autoencoders with Gaussian-process structure.

MICROSOFT RESEARCH CAMBRIDGE — July 2021 – October 2021
Research Intern

- Molecular simulation with deep learning under Dr. Sebastian Nowozin.

CAMBRIDGE UNIVERSITY ENGINEERING DEPARTMENT — 2018 – 2021
Undergraduate Supervisor (analogous to US teaching assistant)

- Small-group supervision for third-year modules 3F7 Information Theory and 3F8 Inference.

CAMBRIDGE UNIVERSITY ENGINEERING DEPARTMENT — Summer 2016
Undergraduate Research Opportunities Program

- Research with Professor Robin Langley on theoretical statistical mechanics. Implemented hard-sphere molecular dynamics simulations in MATLAB.

EDUCATION

PH.D. IN ENGINEERING, UNIVERSITY OF CAMBRIDGE — October 2018 – November 2022
Thesis: *Approximate Inference in Bayesian Neural Networks and Translation Equivariant Neural Processes*.

Doctoral research in machine learning under Professor Richard E. Turner in the Cambridge Machine Learning Group (Computational and Biological Learning Lab). Focus: Bayesian deep learning, approximate inference, deep generative models, meta-learning, equivariances, PAC-Bayes.

M.ENG & B.A. IN ENGINEERING, UNIVERSITY OF CAMBRIDGE October 2014 – June 2018

First Class Honors with Distinction

Specialized in information and computer engineering; Master's project on approximate inference and information theory. Ranked in the top first to second percentile (of ∼300) for the first three years; awarded the Institution of Engineering and Technology Prize in 4th year.

RESEARCH PAPERS

Asterisks denote co-first authorship.

1. Federico Mastroleo, Mariana Borras-Osorio, Shiv Patel, Sarah Peterson, Renthony Wilson, Mohammad Javad Namazi, Mi Zhou, Satomi Shiraishi, ANDREW Y. K. FOONG, David Routman, Mark R Waddle. *From BERT to GPT-4: A systematic review of AI-Driven toxicity extraction and grading in radiation oncology.* Technical Innovations & Patient Support in Radiation Oncology (2025): 100372.
2. Federico Mastroleo, Mariana Borras-Osorio, Shiv Patel, Sarah Peterson, Renthony Wilson, Mi Zhou, Satomi Shiraishi, ANDREW Y. K. FOONG, David Routman, Mark R. Waddle. *Large language models for toxicity extraction in oncology trials: A real-world benchmark in prostate radiotherapy*. Radiotherapy and Oncology (2025): 111348.
3. Sarah Lewis*, Tim Hempel*, Jos Jiménez-Luna*, Michael Gastegger*, Yu Xie*, ANDREW Y. K. FOONG*, Victor Garcia Satorras*, Osama Abdin*, Bastiaan S. Veeling*, Iryna Zaporozhets, Yaoyi Chen, Soojung Yang, Arne Schneuing, Jigyasa Nigam, Federico Barbero, Vincent Stimper, Andrew Campbell, Jason Yim, Marten Lienen, Yu Shi, Shuxin Zheng, Hannes Schulz, Usman Munir, Cecilia Clementi, Frank Noé. *Scalable emulation of protein equilibrium ensembles with generative deep learning*. Science 389, no. 6761 (2025): eadv9817.
4. Richard E. Turner, Cristiana-Diana Diaconu, Stratis Markou, Aliaksandra Shysheya, ANDREW Y. K. FOONG, Bruno Młodozeniec. *Denoising Diffusion Probabilistic Models in Six Simple Steps*. arXiv preprint 2402.04384.
5. Jason Yim, Andrew Campbell, Emile Mathieu, ANDREW Y. K. FOONG, Michael Gastegger, Jos Jimenez-Luna, Sarah Lewis, Victor Garcia Satorras, Bastiaan S. Veeling, Frank Noé, Regina Barzilay, Tommi Jaakkola. *Improved motif-scaffolding with SE(3) flow matching*. Transactions on Machine Learning Research, 07/2024.
6. Jason Yim, Andrew Campbell, ANDREW Y. K. FOONG, Michael Gastegger, Jos Jimenez-Luna, Sarah Lewis, Victor Garcia Satorras, Bastiaan S. Veeling, Regina Barzilay, Tommi Jaakkola, Frank Noé. *Fast protein backbone generation with SE(3) flow matching*. In Machine Learning for Structural Biology Workshop, 2023.
7. Leon Klein*, ANDREW Y. K. FOONG*, Tor Erlend Fjelde*, Bruno Młodozeniec*, Marc Brockschmidt, Sebastian Nowozin, Frank Noé, Ryota Tomioka. *Timewarp: Transferable Acceleration of Molecular Dynamics by Learning Time-Coarsened Dynamics*. In Neural Information Processing Systems, 2023. (Spotlight)
8. Wessel P. Bruinsma*, Stratis Markou*, James Requiema*, ANDREW Y. K. FOONG*, Tom R. Andersson, Anna Vaughan, Anthony Buonomo, J. Scott Hosking, Richard E. Turner. *Autoregressive Conditional Neural Processes*. In International Conference on Learning Representations, 2023.
9. ANDREW Y. K. FOONG*, Wessel P. Bruinsma*, David R. Burt, Richard E. Turner. *How Tight Can PAC-Bayes be in the Small Data Regime?* In Neural Information Processing Systems, 2021.
10. Marcin B. Tomczak, Siddharth Swaroop, ANDREW Y. K. FOONG, Richard E. Turner. *Collapsed Variational Bounds for Bayesian Neural Networks*. In Neural Information Processing Systems, 2021.
11. Wessel P. Bruinsma, James Requeima, ANDREW Y. K. FOONG, Jonathan Gordon, Richard E. Turner. *The Gaussian Neural Process*. In Symposium on Advances in Approximate Bayesian Inference, 2021.
12. ANDREW Y. K. FOONG*, Wessel P. Bruinsma*, Jonathan Gordon*, Yann Dubois, James Requeima, Richard E. Turner. *Meta-Learning Stationary Stochastic Process Prediction with Convolutional Neural Processes*. In Neural Information Processing Systems, 2020. ACCOMPANYING BLOG: `https://yanndubs.github.io/Neural-Process-Family`

13. ANDREW Y. K. FOONG*, David R. Burt*, Yingzhen Li, Richard E. Turner. *On the Expressiveness of Approximate Inference in Bayesian Neural Networks*. In Neural Information Processing Systems, 2020.
14. Tim Pearce, ANDREW Y. K. FOONG, Alexandra Brintrup. *Structured Weight Priors for Convolutional Neural Networks*. In Uncertainty in Deep Learning Workshop, ICML, 2020.
15. ANDREW Y. K. FOONG*, David R. Burt*, Yingzhen Li, Richard E. Turner. *Pathologies of Factorised Gaussian and MC Dropout Posteriors in Bayesian Neural Networks*. In Bayesian Deep Learning Workshop, NEURIPS, 2019.
16. Jonathan Gordon*, Wessel P. Bruinsma*, ANDREW Y. K. FOONG, James Requeima, Yann Dubois, Richard E. Turner. *Convolutional Conditional Neural Processes*. In International Conference on Learning Representations, 2020. (Oral presentation)
17. ANDREW Y. K. FOONG, Yingzhen Li, José Miguel Hernández-Lobato, Richard E. Turner. *'In-Between' Uncertainty in Bayesian Neural Networks*. In Uncertainty in Deep Learning Workshop, ICML, 2019. (Oral presentation)

ORGANISATION AND REVIEWING

- Organiser for the NEURIPS 2021 Approximate Inference in Bayesian Deep Learning Competition.
- Reviewer for the Journal of Machine Learning Research (JMLR), NEURIPS, AISTATS, ICML Uncertainty in Deep Learning Workshop, NEURIPS Bayesian Deep Learning Workshop.

AWARDS AND SCHOLARSHIPS

Trinity Hall Research Studentship 2018–2021
PH.D. Funding.

George and Lilian Schiff Foundation Studentship 2018–2021
PH.D. Funding.

Institution of Engineering and Technology Prize 2018
For outstanding students who have completed an IET accredited course.

Institution of Civil Engineers Baker Prize 2017
Awarded to the two highest scoring students in the Cambridge third-year Engineering course.

BP First Year Prize 2015
Awarded to the four highest scoring students in the Cambridge first-year Engineering course.

TEACHING

AIHC 5100: AI Fundamentals for Healthcare Professionals 09/2025–12/2025
Created a 24-lecture, two-credit course for the Mayo Clinic Graduate School of Biomedical Sciences AI in Healthcare master's program, covering deep learning, including mathematical foundations and modern applications.

REFERENCES

Available upon request.